

REQUIREMENT SPECIFICATION DOCUMENT

E-Commerce Order Management Database System

Week 1 | Database Management Systems

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SUBJECT : DATABASE MANAGAEMENT SYSTEM
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TOPIC : E-Commerce Furniture Management System

1.Introduction

This document lists all the requirements for an E-Commerce Furniture Management System. The purpose is to understand what the system should do before designing the website. This system is similar to online furniture shopping websites where customers can browse, order, and review furniture products.

2.Scope

The E-Commerce Furniture Management System helps customers register, browse furniture, add products to the cart, place orders, make payments, track orders, and give reviews. It also allows the admin to manage products, categories, customers, orders, inventory, and payments.

3.Problem Statement

The traditional furniture shopping process is slow and difficult because customers must visit stores to buy furniture, and shop owners manage products and orders manually. The E-Commerce Furniture Management System provides an online platform to browse furniture, place orders, make payments, and manage products efficiently.

4.Stakeholders

Stakeholder	Responsibility
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Customer	Browses furniture, places orders, makes payments, and writes reviews.
Admin	Manages furniture products, categories, and customer accounts.
Warehouse Staff	Update furniture availability and updates shipping status.
Delivery Team	Delivers orders and updates the delivery status .

5.Functional requirement

➤ Customer Management

- Store customer name, email, phone number, and address.
- Each customer must have a unique Customer ID.
- A customer can update profile details.
- A customer can save more than one delivery address.

➤ Furniture Management

- Store furniture name, description, material, price, and stock quantity
- Furniture must be grouped into categories such as Sofa, Chair, Table, Bed, and Cupboard.
- The system must track stock availability.

➤ Order Management

- A customer can place one or more furniture orders.
- Each order must have a unique Order ID.
- Store order date, total amount, and order status.
- Order status: Pending, Confirmed, Shipped, Delivered, Cancelled.

➤ Payment Management

- Store payment amount, method, date, and status.
- Payment methods: UPI, Credit Card, Debit Card, Net Banking, Cash on Delivery.
- Payment status: Paid, Pending, Failed, Refunded.

➤ Delivery Management

- Store delivery address and expected delivery date.
- Update delivery status.
- Assign delivery staff for each order.

➤ Review Management

- Customers can rate furniture products from 1 to 5 stars.
- Customers can write reviews after delivery.

6.Non-Functional Requirements

➤ Performance

- The website should load quickly.
- Product search results should appear within a few seconds.
- The system should support multiple users at the same time.

➤ Usability

- The website should be simple and easy to use.
- Users should easily browse products and place orders.
- The website should work properly on desktop and mobile devices.

➤ Security

- Customer passwords should be stored securely.
- Payment information should be protected.
- Only administrators can manage products and orders.

7.System Modules

- Customer Management
- Furniture Product Management
- Category Management
- Shopping Cart Management
- Order Management
- Payment Management
- Inventory Management
- Review and Rating Management
- Admin Management

8.Input Requirements

- The system requires the following inputs from users:
- Customer registration details (Name, Email, Phone Number, Address)
- Login credentials (Email and Password)
- Furniture details (Product Name, Category, Price, Stock)
- Customer order details (Selected Products, Quantity, Delivery Address)
- Payment details (Payment Method and Amount)
- Product review and rating (Rating and Review Comment)
- Admin inputs for adding, updating, or deleting furniture products

9.Output Requirements

- The system provides the following outputs:
- Successful registration and login confirmation

- Furniture product list with details
- Shopping cart summary
- Order confirmation with Order ID
- Payment confirmation and payment status
- Order tracking and delivery status
- Customer order history
- Product reviews and ratings
- Admin reports on products, orders, inventory, and payments

10.Constraints

- The website requires an internet connection.
- Only registered users can place orders.
- Real payment gateway integration is not included.
- Live GPS tracking is not supported.
- The project is developed for educational purposes.
- The system supports only online skincare and cosmetics shopping.